Statistical methods for computer science

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Interval Estimation

Intro

- Statistical inference consists in using sample data to estimate unknown parameters upon which the distribution of the characteristic of interest in the population depends
- As any inductive procedure, statistical inference leads to uncertain conclusions
- In general, the point estimate obtained from a sample does not coincide with the true parameter value (even though a good estimate should not be that far)
- The goal of statistical inference is indeed that of quantifying uncertainty associated to the inductive process
- And this is essential for a correct interpretation of a parameter estimate

- In detail, here the goal is that of quantifying uncertainty (deriving from sampling variability) associated to an estimate
- Indeed, its specific value would have quite likely been different had a different sample been observed
- A way for doing this is associating to the point estimate an interval, known as confidence interval, within which the true and unknown parameter values lies with a given probability

Interval estimates or confidence intervals

An interval estimate (or confidence interval) for a parameter θ is the shortest interval, depending on the (random) sample $\boldsymbol{Y} = (Y_1, \ldots, Y_n)'$, that encloses the true θ with a given probability $(1 - \alpha)$. Formally,

$$\text{CI} = [L_1(\boldsymbol{Y}); L_2(\boldsymbol{Y})] \quad : \quad \Pr(L_1(\boldsymbol{Y}) \leq \theta \leq L_2(\boldsymbol{Y})) = 1 - \alpha$$

and

$$L_2(\boldsymbol{Y}) \quad - \quad L_1(\boldsymbol{Y}) = \min$$

The probability $1 - \alpha$ is known as confidence level

To find confidence intervals, we apply the pivot method; this starts from the sampling distribution of a statistic that is a good estimator of the unknown parameter θ

Pivot or pivotal quantity

A pivot (or pivotal quantity), $Q(\theta, \boldsymbol{Y})$, is a quantity such that

- it depends on θ
- it depends on the random sample data $\boldsymbol{Y} = (Y_1, \ldots, Y_n)'$
- it does not depend on other unknown parameters
- its distribution is completely known, it does not depend on θ nor any other unknown parameter
- it is invertible on θ

Pivot method

1. Find a pivot for θ
2. Find the shortest interval for the pivot $Q(\theta, \boldsymbol{Y})$, say $[q_1, q_2]$, s.t.

$$\Pr(q_1 \leq Q(\theta, \boldsymbol{Y}) \leq q_2) = 1 - \alpha$$

   and

$$q_2 - q_1 = \min$$

2. Invert the the inequality wrt θ, so that to obtain

$$L_1(\boldsymbol{Y}) \leq \theta \leq L_2(\boldsymbol{Y})$$

$[L_1(Y); L_2(\boldsymbol{Y})$ is the searched confidence interval

Now, the question is

how do we find a pivotal quantity?

CI for μ in a Gaussian population, with known σ^2

- Let $Y \sim N(\mu, \sigma^2)$ and let us assume σ^2 is known
- A good (ML) estimator for μ is

$$\bar{Y} = \frac{1}{n} \sum_i Y_i$$

- We know that

$$\bar{Y} \sim N(\mu, \sigma^2/n)$$

- However, $\bar{Y}$ is not a pivot as it does not depend on the unknown parameter μ

1. **Find a pivot for μ**

We can standardize $\bar{Y}$

$$Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}}$$

- ✓ it depends on μ
- ✓ it depends on the random sample data $\boldsymbol{Y} = (Y_1, \ldots, Y_n)'$ in $\bar{Y}$
- ✓ it does not depend on other unknown parameters
- ✓ its distribution is completely known (the standard Gaussian), it does not depend on μ nor any other unknown parameter
- ✓ it is invertible on μ

Z is a pivot for μ

2. **Find the shortest interval for the pivot at $(1 - \alpha)$**

- Since the standard Gaussian distribution is symmetric around zero, the shortest interval is the symmetric one around zero
- Denoting by $z_{\alpha/2}$ the standard Gaussian quantile, leaving a probability equal to a $\alpha/2$ on its right, the shortest interval for the pivot is such that

$$\Pr\left(-z_{\alpha/2} \leq \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

3. **Invert the inequality wrt to μ**

$$\Pr\left(-z_{\alpha/2} < \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1 - \alpha$$

$$\Pr\left(-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \bar{Y} - \mu < z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

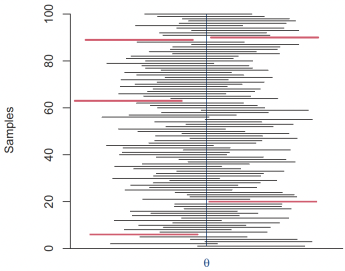
$$\Pr\left(z_{\alpha/2}\frac{\sigma}{\sqrt{n}} > \mu - \bar{Y} > -z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$$\Pr\left(\bar{Y} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{Y} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

That is, a $(1-\alpha)\%$ interval estimator for μ is

$$[\bar{Y} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}; \bar{Y} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]$$

The r.v. $\bar{Y}$ produces intervals that include μ with a given probability $(1-\alpha)$. In this figure, many CIs do include μ, few CIs do not, how many CIs would you expect to not cover μ?



A given sample generates **just one** $(1-\alpha)\%$ interval estimate for the parameter μ

$$[\bar{y} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{y} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}],$$

Interval estimators and estimates for μ in a Gaussian population

The interval estimate for μ is

$$[\bar{y} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{y} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}],$$

while the corresponding estimator is

$$[\bar{Y} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}; \bar{Y} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]$$

The true parameter μ will be included in this interval in the $(1-\alpha)\%$ of the possible samples

- The quantity $z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ is known as margin of error and quantifies the uncertainty associated to the point estimate of μ

- As the confidence level $(1-\alpha)$ grows, α decreases and $z_{\alpha/2}$ increases, so that the confidence interval gets wider

- As n grows, the standard error for $\bar{Y}$ (i.e., $\sigma/\sqrt{n}$) decreases, so the confidence interval gets narrower

Example

We take the weight of an object 5 times

y	3.142	3.163	3.155	3.150	3.141

$\bar{y} = 3.1502$

We assume that the observed weight, due to measurement errors, is a random variable $Y \sim N(\mu, \sigma = 0.01)$

Let's calculate the 95% CI

$$\mathrm{CI} : \bar{y} \pm 1.96 * 0.01/\sqrt{5} = [3.1493, 3.1510]$$

We may be confident (at the $(1-\alpha)\%$) that the mean weight of the object lies in between this interval

CI for μ in a Gaussian population with unknown σ^2

Remember that a Student-t r.v. can be defined as

$$\frac{Z}{\sqrt{\frac{\chi^2_{n-1}}{n-1}}} \sim t_{n-1}$$

with $Z \sim N(0,1)$, and that

$$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}}$$

- ✓ it depends on μ
- ✓ it depends on the random sample data $\boldsymbol{Y} = (Y_1, \ldots, Y_n)'$ in $\bar{Y}$
- ✗ it does not depend on other unknown parameters $\rightarrow$ it depends on σ which is unknown)

1. **Find a pivot for μ**

Since we do not know σ^2, we can use the *unbiased* estimator

$$S^2 = \frac{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}{n-1}$$

Then, let us consider the following random variable

$$\begin{aligned} T = \frac{\bar{Y}-\mu}{S/\sqrt{n}} &= \frac{\bar{Y}-\mu}{S/\sqrt{n}}\sqrt{\frac{\sigma^2(n-1)}{\sigma^2(n-1)}} \\ &= \frac{\bar{Y}-\mu}{\sigma/\sqrt{n}}\sqrt{\frac{\sigma^2(n-1)}{S^2(n-1)}} \\ &= \frac{\bar{Y}-\mu}{\sigma/\sqrt{n}}\Big/\sqrt{\frac{S^2(n-1)}{\sigma^2(n-1)}} \\ &= N(0,1)\Big/\sqrt{\chi^2_{n-1}/(n-1)} \sim t_{n-1} \end{aligned}$$

$$T = \frac{\bar{Y} - \mu}{S/\sqrt{n}}$$

✓ it depends on μ

✓ it depends on the random sample data $\boldsymbol{Y} = (Y_1, \ldots, Y_n)'$ in $\bar{Y}$

✓ it does not depend on other unknown parameters

✓ its distribution is completely known (the Student t distribution, with $n-1$ df), it does not depend on μ nor any other unknown parameter

✓ it is invertible on μ

T is a pivot for μ

2. Find the shortest interval for the pivot at $(1-\alpha)$

- Since the standard Student t distribution is symmetric around zero, the shortest interval is the symmetric one around zero
- Denoting by $t_{\alpha/2,n-1}$ the quantile of the Student t distribution with $(n-1)$ df, leaving a probability equal to a $\alpha/2$ on its right, the shortest interval for the pivot is such that

$$\Pr\left(-t_{\alpha/2,n-1} \leq \frac{\bar{Y}-\mu}{S/\sqrt{n}} \leq t_{\alpha/2,n-1}\right) = 1-\alpha$$

3. **Invert the inequality wrt to μ**

Proceeding as before, we have that

$$P\left(\bar{Y} - t_{\frac{\alpha}{2},n-1}\frac{S}{\sqrt{n}} < \mu < \bar{Y} + t_{\frac{\alpha}{2},n-1}\frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

The $(1-\alpha)\%$ interval estimator for μ is

$$[\bar{Y} - t_{\frac{\alpha}{2},n-1}\frac{S}{\sqrt{n}}; \bar{Y} + t_{\frac{\alpha}{2},n-1}\frac{S}{\sqrt{n}}]$$

Clearly, we observe only one sample, for which $\bar{Y} = \bar{y}$ and $S = s$; the interval estimate at the $(1-\alpha)\%$ for μ is

$$[\bar{y} - t_{\frac{\alpha}{2},n-1}\frac{s}{\sqrt{n}}; \bar{y} + t_{\frac{\alpha}{2},n-1}\frac{s}{\sqrt{n}}]$$

The true parameter μ will be included in this interval in the $(1-\alpha)\%$ of the possible samples

Example

Going back to the measurement error example

- first calculate $s = 0.00920326$
- then look at the distribution tables $t_{0.025,4} = -2.776$, finally

$$\bar{x} \pm t_{\frac{\alpha}{2},n-1} \frac{s}{\sqrt{n}} = [3.1387, 3.1616]$$

CI for σ^2 for a Gaussian population

1. **Find a pivot for μ**

Let us consider the following random variable

$$C = (n-1)\frac{S^2}{\sigma^2} \tag{1}$$

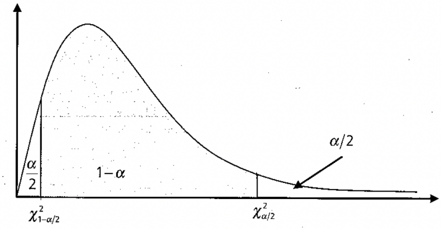
- ✓ it depends on σ^2
- ✓ it depends on the random sample data $\boldsymbol{Y} = (Y_1, \ldots, Y_n)'$ in S
- ✓ it does not depend on other unknown parameters
- ✓ its distribution is completely known ($\chi^2_{(n-1)}$, with $n-1$ df), it does not depend on σ^2 nor any other unknown parameter
- ✓ it is invertible on σ^2

C is a pivot for σ^2

2. **Find the shortest interval for the pivot at $(1-\alpha)$**

- The Chi-square distribution is asymmetric finding the shortest interval is slightly more difficult than before
- Denoting by $\chi^2_{\alpha/2,n-1}$ and $\chi^2_{1-\alpha/2,n-1}$ the quantiles of the chi-square distribution with $(n-1)$ df, leaving a probability equal to a $\alpha/2$ and $1-\alpha/2$ on their right respectively, the shortest interval for the pivot is such that

$$\Pr\left(\chi^2_{1-\frac{\alpha}{2},n-1} < (n-1)\frac{S^2}{\sigma^2} < \chi^2_{\frac{\alpha}{2},n-1}\right) = 1-\alpha$$



3. **Invert the inequality wrt σ^2**

$$1-\alpha = \Pr\left(\frac{\chi^2_{1-\frac{\alpha}{2},n-1}}{(n-1)S^2} < \frac{1}{\sigma^2} < \frac{\chi^2_{\frac{\alpha}{2},n-1}}{(n-1)S^2}\right)$$
$$= \Pr\left(\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2},n-1}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2},n-1}}\right)$$

The $(1-\alpha)\%$ interval estimator for σ^2 is

$$\left[\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2},n-1}}; \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2},n-1}}\right]$$

Clearly, we observe only one sample, for which $S^2 = s^2$; the interval estimate at the $(1-\alpha)\%$ for σ^2 is

$$\left[\frac{(n-1)s^2}{\chi_{\frac{\alpha}{2},n-1}}, \frac{(n-1)s^2}{\chi_{1-\frac{\alpha}{2},n-1}}\right]$$

The true parameter σ^2 will be included in this interval in the $(1-\alpha)\%$ of the possible samples

CI for p in a Bernoulli population

- Let $Y \sim Be(p)$
- A good (ML) estimator for p is

$$\hat{P} = \bar{Y} = \frac{1}{n} \sum_i Y_i$$

- We know that, for n sufficiently large,

$$\hat{P} \overset{a}{\sim} N\left(p, \frac{p(1-p)}{n}\right)$$

- However, $\hat{P}$ is not a pivot as it does not depend on the unknown parameter p

1. **Find a pivot for p**

We can standardize $\hat{P}$

$$Z = \frac{\hat{P} - p}{\sqrt{p(1-p)/n}} \overset{a}{\sim} N(0,1)$$

- ✓ it depends on μ
- ✓ it depends on the random sample data $\boldsymbol{Y} = (Y_1, \ldots, Y_n)'$ in $\bar{Y}$
- ✓ it does not depend on other unknown parameters
- ✓ its distribution is completely known (the standard Gaussian), it does not depend on μ nor any other unknown parameter
- ✓ it is invertible on μ

Z is a pivot for p

- It is however worth noticing that both the numerator and the denominator of Z depend on p
- This leads to a quadratic inequality on p when inverting wrt to this parameter
- To solve the issue, we can substitute the standard error for $\hat{P}$ at the denominator of the pivotal quantity by the corresponding estimator
- This leads to the following pivot

$$Z = \frac{\hat{P} - p}{\sqrt{\hat{P}(1 - \hat{P})/n}} \overset{a}{\sim} N(0, 1)$$

2. **Find the shortest interval for the pivot at $(1-\alpha)$**

$$\Pr\left(-z_{\frac{\alpha}{2}} < \frac{\hat{P}-p}{\sqrt{\hat{P}(1-\hat{P})/n}} < z_{\frac{\alpha}{2}}\right) \simeq 1-\alpha$$

3. **Invert the inequality wrt *p***

$$\begin{aligned} 1-\alpha &\simeq \Pr\left(-z_{\frac{\alpha}{2}}\sqrt{\hat{P}(1-\hat{P})/n} < \hat{P}-p < z_{\frac{\alpha}{2}}\sqrt{\hat{P}(1-\hat{P})/n}\right) \\ &\simeq \Pr\left(\hat{P}-z_{\frac{\alpha}{2}}\sqrt{\hat{P}(1-\hat{P})/n} < p < \hat{P}+z_{\frac{\alpha}{2}}\sqrt{\hat{P}(1-\hat{P})/n}\right) \end{aligned}$$

Note that this is an approximate confidence interval for p

The $(1-\alpha)\%$ approximate interval estimator for p is

$$\left[\hat{P} - z_{\frac{\alpha}{2}}\sqrt{\hat{P}(1-\hat{P})/n};\, \hat{P} + z_{\frac{\alpha}{2}}\sqrt{\hat{P}(1-\hat{P})/n}\right]$$

while the corresponding interval estimate is

$$\hat{p} - z_{\frac{\alpha}{2}}\sqrt{\hat{p}(1-\hat{p})/n};\, \hat{p} + z_{\frac{\alpha}{2}}\sqrt{\hat{p}(1-\hat{p})/n}$$

Example

- Let $X \sim Be(p)$
- $n = 380$
- $\sum_i y_i = 90$
- Compute the CI at the 99% confidence level for p
  - $\hat{p} = \bar{y} = 90/380 = 0.237$
  - $\widehat{se}(\hat{P}) = \sqrt{\frac{0.237(1-0.237)}{380}} = 0.022$
  - $z_{0.005} = 2.575$
  - $\mathrm{CI} = [0.237 - 2.575 \times 0.022; 0.237 + 2.575 \times 0.022;] = [0.18; 0.294]$

Large-Sample CI for the mean

- We can use the pivotal quantity approach to construct a confidence interval for a population mean, μ of a quantitative variable, using n independent observations, such as in a randomized study
- In practice, we usually do not know the shape of the population distribution, other than what we learn from the sample data
- By the CLT, for large n the sampling distribution of $\bar{Y}$ is approximately normal with mean μ and standard error $\sigma_{\bar{Y}} = \sigma/\sqrt{n}$
- Therefore, the random variable

$$Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

  is a pivot for μ
- The method is designed for large n, and then the sample standard deviation s is close expected to be close to the true σ, so we could rely on this approximate Gaussian distribution even for unknown σ